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DS-B

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HOMEWORK #9

Q1: $A = \begin{pmatrix} 2 & 1 \\ 4 & -1 \end{pmatrix}$

① Eigenvalues: $\lambda^2 - tr(A)\lambda + detA = 0$

$\lambda^2 - 1\lambda - 6 = 0$

$\lambda_1 = 3 \quad , \quad \lambda_2 = -2$

② $\lambda_1 = 3$ Case:

$$A - 3I = \begin{pmatrix} -1 & 1 \\ 4 & -4 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 1 \\ 4 & -4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$A\left[\begin{array}{cc|c} -1 & 1 & 0 \\ 4 & -4 & 0 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 4 & -4 & 0 \end{array}\right] \begin{matrix} \\ -R1 \\ R2-4R1 \end{matrix} \sim \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right].$$

$x - y = 0$

$x = y$

$\vec{v_1} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ x \end{pmatrix} = x\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

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$\lambda_2 = -2$ Case:

$$A + 2I = \begin{pmatrix} 4 & 1 \\ 4 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 1 \\ 4 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\left[\begin{array}{cc|c} \textcircled{4} & 1 & 0 \\ 4 & 1 & 0 \end{array}\right] \sim \left[\begin{array}{cc|c} \textcircled{4} & 1 & 0 \\ 0 & 0 & 0 \end{array}\right] \; R2 - R1$$

$4x + y = 0$

$y = -4x$

$$\vec{v_2} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ -4x \end{pmatrix} = x\begin{pmatrix} 1 \\ -4 \end{pmatrix}$$

(a) $\begin{pmatrix} 2 & 1 \\ 4 & -1 \end{pmatrix}^{10}\begin{pmatrix} 5 \\ 1 \end{pmatrix}$

$A = PDP^{-1}$

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -4 \end{pmatrix}, \; D = \begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}, \; P^{-1} = \begin{pmatrix} +4/5 & +1/5 \\ +1/5 & -1/5 \end{pmatrix}$$

$$(A)^{10} = \begin{pmatrix} 1 & 1 \\ 1 & -4 \end{pmatrix}\begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}^{10}\begin{pmatrix} 4/5 & 1/5 \\ 1/5 & -1/5 \end{pmatrix}\begin{pmatrix} 5 \\ 1 \end{pmatrix}$$

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$$\begin{pmatrix} 1 & 1 \\ 1 & -4 \end{pmatrix}\begin{pmatrix} 59049 & 0 \\ 0 & 1024 \end{pmatrix}\begin{pmatrix} 4/5 & 1/5 \\ 1/5 & -1/5 \end{pmatrix}\begin{pmatrix} 5 \\ 1 \end{pmatrix}$$ Moved

$$\begin{pmatrix} (1)(59049)+(1)(0) & (1)(0)+(1)(1024) \\ (1)(59049)+(-4)(0) & (1)(0)+(-4)(1024) \end{pmatrix}\begin{pmatrix} (4/5)(5)+(1/5)(1) \\ (1/5)(5)+(-1/5)(1) \end{pmatrix}$$

$$\begin{pmatrix} 59049 & 1024 \\ 59049 & -4096 \end{pmatrix}\begin{pmatrix} 4.2 \\ 0.8 \end{pmatrix} = \begin{pmatrix} (59049)(4.2)+(1024)(0.8) \\ (59049)(4.2)+(-4096)(0.8) \end{pmatrix}$$

$$= \begin{pmatrix} 248825 \\ 244729 \end{pmatrix}$$

(b) Diagonalization: $PDP^{-1} = A$

$$\begin{pmatrix} 1 & 1 \\ 1 & -4 \end{pmatrix}\begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}\begin{pmatrix} 4/5 & 1/5 \\ 1/5 & -1/5 \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ 3 & 8 \end{pmatrix}\begin{pmatrix} 4/5 & 1/5 \\ 1/5 & -1/5 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 4 & -1 \end{pmatrix}$$

(c) $A^{10} = PD^{10}P^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & -4 \end{pmatrix}\begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}^{10}\begin{pmatrix} 4/5 & 1/5 \\ 1/5 & -1/5 \end{pmatrix}$

$$A^{10} = \begin{pmatrix} 1 & 1 \\ 1 & -4 \end{pmatrix}\begin{pmatrix} 59049 & 0 \\ 0 & 1024 \end{pmatrix}\begin{pmatrix} 4/5 & 1/5 \\ 1/5 & -1/5 \end{pmatrix} = \begin{pmatrix} 47444 & 11605 \\ 46420 & 12629 \end{pmatrix}$$

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